

Naive Bayes Classifier Intuition

Toss	Venue	Outlook	Result
won	mumbai	overcast	won
lost	chennai	sunny	won
won	kolkata	sunny	won
won	chennai	sunny	won
lost	mumbai	sunny	lost
won	chennai	overcast	lost
won	kolkata	overcast	lost
won	chennai	sunny	won

From given input conditions whether chennai will win or lose a match.

(Given)

⇒ Input = {lost, Mumbai, Sunny}

From this input we are going to

predict whether chennai will loose or win.

Using Naive Bayes $P(A|B) = \frac{P(B|A) P(A)}{P(B)}$

Given $\Rightarrow \{ \text{lost} \cap \text{Mumbai} \cap \text{Sunny} \}$

A

B

$$P(W | \text{lost} \cap \text{Mumbai} \cap \text{Sunny})$$

$$= \frac{P(\text{lost}, \text{Mumbai}, \text{Sunny} | W) P(W)}{P(\text{lost}, \text{Mumbai}, \text{Sunny})}$$

$$P(L | \text{Lost}, \text{Mumbai}, \text{Sunny})$$

$$= \frac{P(\text{lost}, \text{Mumbai}, \text{Sunny} | L) P(L)}{P(\text{lost}, \text{Mumbai}, \text{Sunny})}$$

Result = $P(L)$ or $P(W)$ whichever is greater

g

Denominator is same in both terms

$$\begin{aligned}\therefore P(W | \text{lost, mumbai, sunny}) &= P(\text{lost, mumbai, sunny} | W) P(W) \\ &= P(\text{lost} | W) P(\text{mumbai} | W) \\ &\quad P(\text{sunny} | W) \cdot P(W)\end{aligned}$$

$$= \frac{1}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{5}{8}$$

$$\begin{aligned}P(L | \text{Lost, mumbai, Sunny}) &= P(\text{lost, mumbai, sunny} | L) P(L) \\ &= P(\text{lost} | L) P(\text{mumbai} | L) P(\text{sunny} | L) P(L) \\ &= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{3}{8}\end{aligned}$$